Every general model over actual F4 and F9

Lean proves point-count classifications and modular embedding-degree bounds for every nonsingular general Weierstrass model over explicit actual fields with 4 and 9 elements. The source criterion is exactly $\gcd(t,q)\neq 1$ with q the proved field cardinality and $t=q+1-\#E$ the actual point-group trace. No general Hasse or Waterhouse classification premise is assumed for these checked fields. [pcc:unverified]

Field	General coefficient tuples covered	Source-supersingular point counts
F4	1,024	1, 3, 5, 7, 9
F9	59,049	4, 7, 10, 13, 16

Every prime divisor r has modular degree $\mathrm{ord}_r(q)\leq 6$, coprimality with q and an actual point of order r. Concrete prime-order-7/13 curves over F4/F9 both have exact modular degree 3. [pcc:unverified]

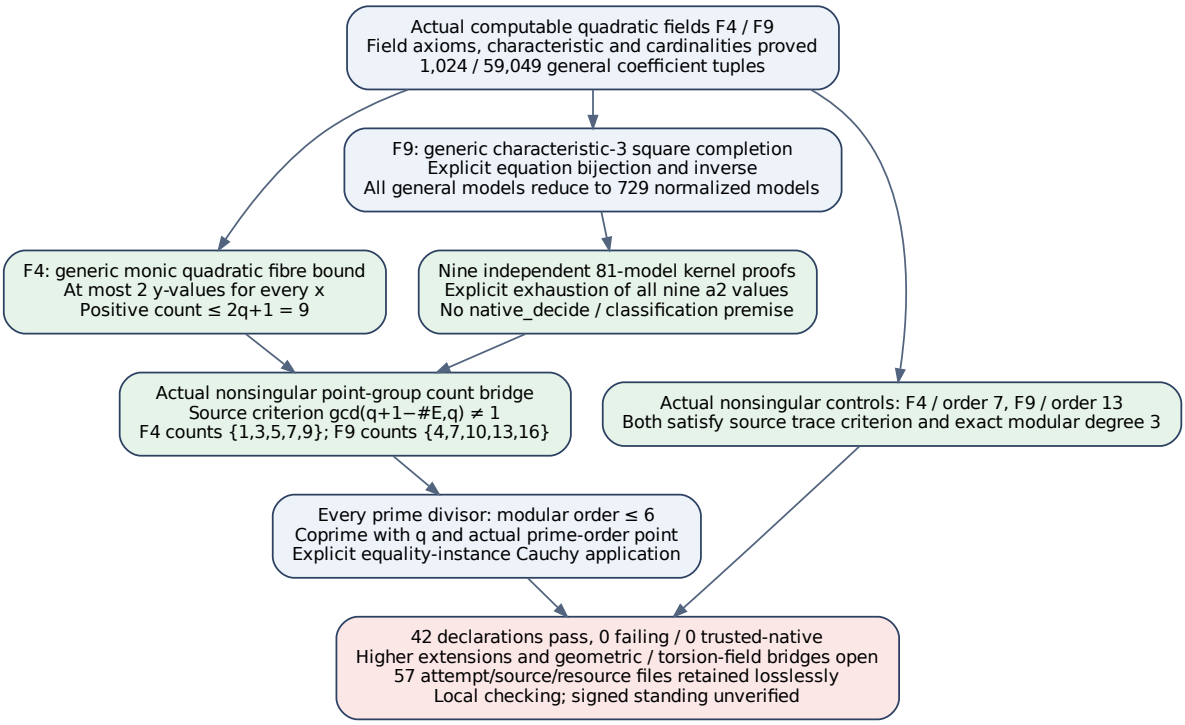

Figure 1: Complete model coverage through actual fields, bijections and kernel checks. [pcc:unverified]

Proof obligations, controls and assurance

F4 is QuadraticAlgebra (ZMod 2) 1 1, with $\alpha^2=\alpha+1$. F9 is QuadraticAlgebra (ZMod 3) (−1) 0, with $\beta^2=-1$. Root-free base polynomials, field axioms, computable arithmetic, characteristics and cardinalities are proved. Composite ZMod 4/9 rings are not used as fields. [pcc:unverified]

For F4, a monic quadratic fibre has at most two roots. An explicit sigma-fibre equivalence gives counted(E)≤2q+1=9. Positivity and the source trace-gcd criterion yield every count case. This is a structural all-model proof, not a claimed 1,024-row native enumeration. [pcc:unverified]

For F9, Lean proves characteristic-3 square completion on every general model, with $y'=y-a_1x-a_3$ and coefficients $a_2'=a_2+a_1^2$, $a_4'=a_4-a_1a_3$, $a_6'=a_6+a_3^2$. Equation equivalence, an explicit coordinate map and inverse, and preservation of the counted set are checked. All 729 normalized models are classified by nine separate 81-model kernel decisions and explicit a_2 exhaustion. The bijection transfers this to all 59,049 general tuples. [pcc:unverified]

Nonsingularity binds the counted equation set to actual Mathlib points. The checked trace lists imply the prime-divisor modular bound and coprimality. Cauchy's theorem produces an actual prime-order point with the caller's equality decision and group instance kept explicit. The actual controls are $y^2+\alpha y=x^3+\alpha^2x$ over F4, of order 7, and $y^2=x^3+\beta x+1$ over F9, of order 13. Both are nonsingular, satisfy the exact source criterion and have modular degree 3. [pcc:unverified]

Module group	Checked obligation
QuadraticModels / SourceTrace	Actual fields, cardinalities, populations and exact source criterion
FibreBound / CompleteSquare	Generic quadratic root bound and characteristic-3 count bijection
NormalizedCase00–22 / NormalizedAll	All 729 normalized models with exact kernel decisions
QuaternaryClassification / NonaryClassification	Every general model, actual counts and prime-divisor conclusions
ExtensionControls / QuadraticCertificate	Actual degree-3 controls and combined certificate

The whole audit passes 42 declarations, zero failures and zero trusted-native exceptions. Lean 4.35.0 / Mathlib 56a00585; only propext, Classical.choice and Quot.sound. No holes, custom axioms, native_decide, linter waivers or increased elaboration limits. All 57 attempt/source/resource files and five gate files have byte-equal archive checks. The first monolithic reduction was resource-interrupted with SIGTERM/143; its source/log/resource snapshot and all rejected attempts are preserved. The case import chain serializes large kernel reductions in normal Lake builds. [pcc:unverified]

FST 2010, printed pages 230–231, and its page-237 trace list, plus Waterhouse 1969 Theorem 4.1, printed pages 536–537, supply the checked source definitions and known classification context. Exact URLs, primary hashes and limited read scopes are retained. No new full-paper read or novelty claim. Higher extensions, geometric supersingularity equivalence and full torsion-field interpretation remain separate obligations. [pcc:unverified]

No native experiment, fresh sampled certificate, timing claim or paid compute. KB wrapper status 127 / MCP unavailable; bounded primary, repository and Mathlib fallback. Conductor unavailable. Source commit, packet and actual verdict are in verification.json; signed standing remains unverified. Proof map and pins: README.md and sources.json. [pcc:unverified]